QFF 2026 Phase 2 — Quantum Simulation Track

PS-S3: Noise Accumulation in Trotter Circuits

StepWise-Q: Optimal Trotter Step Prediction & Hardware-Aware Co-Design

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Executive Summary

- **The Dilemma:** In Trotterized quantum simulation, increasing step count n reduces algorithmic discretization error ($\mathcal{O}(1/n)$), but monotonically inflates gate depth and accumulates physical device noise ($1 - \Lambda^n$). Under a fixed noise model, there exists an optimal balance point n^*.
- **Headline Findings:**
  - **Processor A (Tree Topology):** Lacks cycles $\to$ forces SWAP routing overhead (11 SWAPs, 33 extra CX at $n=6$). Shifts $n^* = 6$ with peak fidelity $F = 0.5774$.
  - **Processor B (Heavy-Hex):** Contains 4-cycle (0 SWAPs for ring), but has node degree 5. Optimal $n^* = 7$ with peak fidelity $F = 0.6628$.
  - **Processor D (2×3 Ladder, 6 Qubits, Native RZZ):** Custom ladder architecture providing two native 4-cycles (e.g. {0,1,4,3}), max degree ≤ 3 (versus B's degree 5), diameter 3, and native R_{ZZ} basis gates. Eliminates 100% of routing SWAPs for the periodic TFIM ring and halves two-qubit entangling gates per Trotter step (4 RZZ vs 8 CX). Unlocks deeper optimal depth $n^* = 8$ with peak fidelity $F = 0.7528$ (+9.0 percentage points higher than Processor B, and +17.5 percentage points higher than Processor A), while cutting farthest Bell-state pair distance to 3 hops (7 CXs vs 10 CXs on a 3×3 grid).
  - **Noise Sweep on D:** As noise scale increases ($0.5\times \to 1.0\times \to 2.0\times$), the optimal step count shifts leftward: $n^* = 8 \to 8 \to 7$.
  - **Bonus Error Mitigation:** Single-qubit readout inversion (factor $(1 - 2p_{\text{readout}}) = 0.98$) and Zero-Noise Extrapolation (ZNE via 2nd-order Richardson extrapolation across noise scales) recover staggered magnetization M_s from raw noisy 0.4949 ± 0.0105 to 0.6224 ± 0.0352 (~81% error reduction toward exact reference 0.6524, within 1σ).

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1. Environment Setup & Execution Control

Set `RECOMPUTE = True` to execute all quantum simulation circuits on Aer, or `False` to rapidly reload and inspect verified tables. All random seeds are explicitly initialized to guarantee 100% numerical reproducibility.

In [1]:
```
# Start execution timer to verify 30-minute competition runtime limit
import time
NOTEBOOK_START_TIME = time.time()

# Explicit Seed Configuration for 100% Numerical Reproducibility
SEED = 42
SHOTS = 4096
RECOMPUTE = True

import os, sys, json, csv
import numpy as np
np.random.seed(SEED)

import matplotlib.pyplot as plt
import qiskit, qiskit_aer
from qiskit.quantum_info import Statevector, state_fidelity

from qff_sim import (
    PROCESSOR_A,
    PROCESSOR_B,
    load_processor,
    validate_processor_d,
    bell_experiment,
    tfim_hamiltonian,
    observable_ops,
    trotter_circuit,
    exact_evolution,
    circuit_metrics,
    noisy_state_fidelity,
    run_z_measurement,
)

print(f'Qiskit Version: {qiskit.__version__} | Aer Version: {qiskit_aer.__ve
print(f'Official Benchmark Settings: SEED={SEED}, SHOTS={SHOTS}')
print(f'Execution Mode: RECOMPUTE={RECOMPUTE}')
os.makedirs('results/tables', exist_ok=True)
os.makedirs('results/figures', exist_ok=True)
```

```
Qiskit Version: 1.4.6 | Aer Version: 0.18.0
Official Benchmark Settings: SEED=42, SHOTS=4096
Execution Mode: RECOMPUTE=True
```

2. Objective 2 — Processor D Proposal & Architectural Validation

Our custom Processor D adopts a **6-qubit 2×3 ladder topology** with 7 edges, max degree 3, and diameter 3. It embeds two native 4-cycles (`0–1–4–3` and `1–2–5–4`) eliminating all SWAP overhead for 4-spin ring TFIM simulation.

In [2]:
```
processor_d_path = 'processors/processor_D.json'
D = load_processor(processor_d_path)

# Validate against official design checker rules for N=4 qubits
val_errors = validate_processor_d(D, 4)
assert len(val_errors) == 0, f'Processor D validation failed: {val_errors}'

print(f"Loaded '{D['name']}': {D['num_qubits']} qubits, {len(D['edges'])} ed
print(f"Basis gates: {D['basis_gates']}")
print(f"Error rates: {D['error_rates']}")
print('Official Checker Status: OK – Processor D satisfies all design rules.
```

```
Loaded 'Processor D': 6 qubits, 7 edges.
Basis gates: ['rz', 'sx', 'x', 'rzz']
Error rates: {'p1': 0.001, 'p2': 0.005, 'p_readout': 0.01}
Official Checker Status: OK – Processor D satisfies all design rules.
```

3. Objective 1 — Bell-State Preparation and Verification

We prepare $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ on Processor D on:

1. **Near Pair:** directly connected qubits $(0, 1)$ ($d = 1$, 1 CX gate).
2. **Far Pair:** maximum distance pair $(0, 5)$ across the 2×3 ladder ($d = 3$, 7 CX gates via $1 + 3(d-1)$).

Each pair is evaluated under ideal and noisy conditions across XX, YY, ZZ bases with 4,096 shots each (12 circuits total).

In [3]:
```
near_pair = (0, 1)  # distance = 1 (1 CX)
far_pair = (0, 5)   # distance = 3 (7 CX via 2 SWAPs)

bell_rows = []
if RECOMPUTE or not os.path.exists('results/tables/bell.csv'):
    for pair in [near_pair, far_pair]:
        for noisy in [False, True]:
            res = bell_experiment(D, pair, noisy=noisy, shots=SHOTS)
            cond_str = 'noisy' if noisy else 'ideal'
            bell_rows.append({
                'pair': f"{pair[0]}-{pair[1]}",
                'pair_distance': res['pair_distance'],
                'condition': cond_str,
                'two_qubit_gates': res.get('cx', 1),
                'XX': res['XX'], 'XX_se': res['XX_se'],
                'YY': res['YY'], 'YY_se': res['YY_se'],
                'ZZ': res['ZZ'], 'ZZ_se': res['ZZ_se'],
                'fidelity': res['fidelity'], 'fidelity_se': res['fidelity_se
            })
```

```
            print(f"  Pair {pair} ({res['pair_distance']} hops) [{cond_str.u

    with open('results/tables/bell.csv', 'w', newline='') as f:
        w = csv.DictWriter(f, fieldnames=['pair', 'pair_distance', 'conditic
        w.writeheader()
        w.writerows(bell_rows)
else:
    with open('results/tables/bell.csv', 'r') as f:
        bell_rows = list(csv.DictReader(f))
    for r in bell_rows:
        print(f"  Pair {r['pair']} ({r['pair_distance']} hops) [{r['conditic
```

```
Pair (0, 1) (1 hops) [IDEAL]: F = 1.0000 ± 0.0000 (2Q gates: 1)
Pair (0, 1) (1 hops) [NOISY]: F = 0.9598 ± 0.0022 (2Q gates: 1)
Pair (0, 5) (3 hops) [IDEAL]: F = 1.0000 ± 0.0000 (2Q gates: 7)
Pair (0, 5) (3 hops) [NOISY]: F = 0.9384 ± 0.0027 (2Q gates: 7)
```

4. Objective 3 — Exact Reference for Ring TFIM

We compute the exact reference state $|\psi_{\text{exact}}(T)\rangle$ and staggered magnetization $M_s(T)$ at $T=2.0$ evolved from Néel $|1010\rangle$ under $H = -\sum Z_i Z_{i+1} - \sum X_i$ via SciPy matrix exponentiation.

In [4]:
```
H_ring = tfim_hamiltonian(4, J=1.0, g=1.0, periodic=True)
psi0 = Statevector(trotter_circuit(4, 1.0, 1.0, 0.0, 0))
target_state = exact_evolution(H_ring, psi0, [2.0])[0]

ops = observable_ops(4)
ms_exact = float(np.real(target_state.expectation_value(ops['ms'])))
print(f'Exact Reference Staggered Magnetization Ms(T=2.0) = {ms_exact:.6f}')
```

```
Exact Reference Staggered Magnetization Ms(T=2.0) = 0.652444
```

5. Objective 3 — Headline Step Scan ($n \in [1..16]$) on Processors A, B, and D

We evaluate noisy density-matrix state fidelity $F(n)$ and 4,096-shot staggered magnetization $M_s(n)$ across step counts $n \in [1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16]$. We also perform the noise sweep on Processor D at $0.5\times, 1.0\times, 2.0\times$ noise scales.

In [5]:
```
steps_list = [1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16]
scan_configs = [
    (PROCESSOR_A, 'A', 1.0),
    (PROCESSOR_B, 'B', 1.0),
    (D, 'D', 0.5),
    (D, 'D', 1.0),
    (D, 'D', 2.0)
]

s3_scan_rows = []
```

```
summary_rows = []
resources_rows = []

def write_table(path, rows, fields):
    with open(path, 'w', newline='') as f:
        w = csv.DictWriter(f, fieldnames=fields)
        w.writeheader()
        w.writerows(rows)

if RECOMPUTE or not os.path.exists('results/tables/s3_scan.csv'):
    pure_fidelities = {}
    metrics_cache = {}

    for n in steps_list:
        qc_pure = trotter_circuit(4, 1.0, 1.0, 2.0, n, periodic=True)
        pure_fidelities[n] = float(state_fidelity(target_state, Statevector(
        for p_obj, p_code in [(PROCESSOR_A, 'A'), (PROCESSOR_B, 'B'), (D, 'D
            zz_mode = 'rzz' if 'rzz' in p_obj.get('basis_gates', []) else 'c
            qc = trotter_circuit(4, 1.0, 1.0, 2.0, n, periodic=True, zz_mode
            m = circuit_metrics(qc, p_obj)
            metrics_cache[(p_code, n)] = m
            resources_rows.append({
                'ps': 'PS-S3', 'circuit_label': f'trotter_o1_n{n}',
                'processor': p_code, 'executable': m['executable'],
                'logical_qubits': m['logical_qubits'], 'physical_qubits': m[
                'logical_2q': m['logical_cx'], 'two_q': m['cx'],
                'depth': m['depth'], 'two_q_depth': m['two_q_depth'],
                'swaps': m['swaps'], 'routing_overhead_2q': m['cx_overhead']
            })

    for proc, p_code, scale in scan_configs:
        best_n, best_f = None, -1.0
        for n in steps_list:
            zz_mode = 'rzz' if 'rzz' in proc.get('basis_gates', []) else 'cx
            qc = trotter_circuit(4, 1.0, 1.0, 2.0, n, periodic=True, zz_mode
            m = metrics_cache[(p_code, n)]
            f_noisy = noisy_state_fidelity(qc, target_state, proc, scale=sca
            zm = run_z_measurement(qc, proc, shots=SHOTS, noisy=True, scale=
            if f_noisy > best_f:
                best_f, best_n = f_noisy, n
            s3_scan_rows.append({
                'processor': p_code, 'noise_scale': scale, 'steps': n,
                'fidelity_noisy': f_noisy, 'fidelity_trotter_ideal': pure_fi
                'ms_noisy': zm['ms'], 'ms_noisy_se': zm['ms_se'], 'ms_exact'
                'logical_2q': m['logical_cx'], 'swap_2q': m['cx_overhead'],
            })
        summary_rows.append({
            'processor': p_code, 'noise_scale': scale,
            'n_star': best_n, 'fidelity_at_n_star': best_f
        })
        print(f'Processor {p_code} (scale {scale}x): Optimal n* = {best_n},

    write_table('results/tables/resources.csv', resources_rows, ['ps', 'circ
    write_table('results/tables/s3_scan.csv', s3_scan_rows, ['processor', 'r
    write_table('results/tables/s3_summary.csv', summary_rows, ['processor',
    print('Scan tables exported successfully.')
```

```
else:
    with open('results/tables/s3_summary.csv', 'r') as f:
        summary_rows = list(csv.DictReader(f))
    with open('results/tables/s3_scan.csv', 'r') as f:
        s3_scan_rows = list(csv.DictReader(f))
    print('Loaded results/tables/s3_summary.csv:')
    for r in summary_rows:
        print(f"  Processor {r['processor']} (scale {r['noise_scale']}x): n*
```

```
Processor A (scale 1.0x): Optimal n* = 6, Peak Fidelity = 0.577419
Processor B (scale 1.0x): Optimal n* = 7, Peak Fidelity = 0.662816
Processor D (scale 0.5x): Optimal n* = 8, Peak Fidelity = 0.841453
Processor D (scale 1.0x): Optimal n* = 8, Peak Fidelity = 0.752817
Processor D (scale 2.0x): Optimal n* = 7, Peak Fidelity = 0.620712
Scan tables exported successfully.
```

6. Bonus — Error Mitigation on Processor D at Dynamically Identified n^*

We read n^* dynamically from `s3_summary.csv` rather than hardcoding it. We apply:

1. **Readout Error Inversion:** Single-qubit observable correction with factor $(1 - 2p_{\text{readout}}) = 0.98$ (NOT 0.98^4, since M_s is a sum of single-qubit Pauli Z terms).
2. **Zero-Noise Extrapolation (ZNE):** 2nd-order Richardson extrapolation across noise scales $\lambda \in [0.5, 1.0, 2.0]$. Since the noise model scales $p_1, p_2,$ and p_{readout} simultaneously, ZNE extrapolates both gate noise and readout attenuation to zero. We do NOT double-count readout correction on top of ZNE.

In [6]:
```
# Dynamically extract n_star for Processor D at 1.0x noise
n_star = [int(r['n_star']) for r in summary_rows if r['processor'] == 'D' ar
print(f'Identified Optimal Step Count for Processor D: n* = {n_star}')

raw_entry = [r for r in s3_scan_rows if r['processor'] == 'D' and float(r['r
ms_raw = float(raw_entry['ms_noisy'])
ms_raw_se = float(raw_entry['ms_noisy_se'])

# 1. Readout Error Mitigation (Single-Qubit Observable Inversion)
p_ro = 0.01
ro_factor = 1.0 - 2.0 * p_ro  # 0.98
ms_ro_mit = ms_raw / ro_factor
ms_ro_se = ms_raw_se / ro_factor

# 2. Zero-Noise Extrapolation (ZNE via Richardson Extrapolation on noise sca
entry_05 = [r for r in s3_scan_rows if r['processor'] == 'D' and float(r['no
entry_20 = [r for r in s3_scan_rows if r['processor'] == 'D' and float(r['no

ms_05, se_05 = float(entry_05['ms_noisy']), float(entry_05['ms_noisy_se'])
ms_10, se_10 = ms_raw, ms_raw_se
ms_20, se_20 = float(entry_20['ms_noisy']), float(entry_20['ms_noisy_se'])
```

```
poly = np.polyfit([0.5, 1.0, 2.0], [ms_05, ms_10, ms_20], 2)
ms_zne = float(poly[2])  # Extrapolated value at lambda = 0

# Analytic error propagation for Richardson extrapolation
w1, w2, w3 = 8.0 / 3.0, -2.0, 1.0 / 3.0
ms_zne_se = float(np.sqrt((w1 * se_05)**2 + (w2 * se_10)**2 + (w3 * se_20)**

mitigation_rows = [
    {'method': 'readout_error_mitigation', 'ms_raw': ms_raw, 'ms_mitigated':
    {'method': 'zero_noise_extrapolation_richardson', 'ms_raw': ms_raw, 'ms_
]
write_table('results/tables/mitigation.csv', mitigation_rows, ['method', 'ms

err_raw = abs(ms_raw - ms_exact)
err_ro = abs(ms_ro_mit - ms_exact)
err_zne = abs(ms_zne - ms_exact)
print(f'Exact Reference Ms:    {ms_exact:.6f}')
print(f'Raw Noisy Ms:          {ms_raw:.6f} ± {ms_raw_se:.6f} (Error: {err_r
print(f'Readout Mitigated Ms:  {ms_ro_mit:.6f} ± {ms_ro_se:.6f} (Error: {err
print(f'ZNE Mitigated Ms:      {ms_zne:.6f} ± {ms_zne_se:.6f} (Error: {err_z
```

```
Identified Optimal Step Count for Processor D: n* = 8
Exact Reference Ms:    0.652444
Raw Noisy Ms:          0.494873 ± 0.010522 (Error: 0.157571)
Readout Mitigated Ms:  0.504972 ± 0.010737 (Error: 0.147472)
ZNE Mitigated Ms:      0.622396 ± 0.035172 (Error: 0.030049, 80.9% reductio
n)
```

7. Results Visualization & Figure Generation

The notebook actively generates and saves all 6 publication figures using Matplotlib.

In [7]:

```
plt.rcParams.update({
    'font.size': 11, 'axes.labelsize': 12, 'axes.titlesize': 13,
    'xtick.labelsize': 10, 'ytick.labelsize': 10, 'legend.fontsize': 10,
    'font.family': 'sans-serif'
})

# --- Figure 1: Processor Topologies ---
fig, axes = plt.subplots(1, 3, figsize=(15, 4.5), dpi=300)
# Processor A
ax = axes[0]
coords_A = {0: (0, 1), 1: (1, 1), 2: (2, 2), 3: (1, 0), 4: (2, -0.5)}
for u, v in PROCESSOR_A['edges']:
    ax.plot([coords_A[u][0], coords_A[v][0]], [coords_A[u][1], coords_A[v][1
for q, (x, y) in coords_A.items():
    ax.scatter(x, y, s=550, color='#6929C4', edgecolor='black', zorder=2)
    ax.text(x, y, str(q), color='white', ha='center', va='center', weight='b
ax.set_title('Processor A (5 Qubits, Tree)\nNo 4-Cycles → High SWAP Overhead
ax.set_xlim(-0.8, 2.8); ax.set_ylim(-1.2, 2.8); ax.axis('off')

# Processor B
ax = axes[1]
coords_B = {0: (0, 0), 1: (1, 1), 2: (2, 1), 3: (3, 0), 4: (2.5, -1), 5: (1.
for u, v in PROCESSOR_B['edges']:
```

```
    ax.plot([coords_B[u][0], coords_B[v][0]], [coords_B[u][1], coords_B[v][1
for q, (x, y) in coords_B.items():
    c = '#DA1E28' if q == 6 else '#0062FF'
    ax.scatter(x, y, s=550, color=c, edgecolor='black', zorder=2)
    ax.text(x, y, str(q), color='white', ha='center', va='center', weight='b
ax.set_title('Processor B (9 Qubits, Heavy-Hex)\nContains 4-Cycle, but Node
ax.set_xlim(-1.2, 3.8); ax.set_ylim(-1.6, 1.8); ax.axis('off')

# Processor D (2x3 ladder)
ax = axes[2]
coords_D = {0: (0, 1), 1: (1, 1), 2: (2, 1), 3: (0, 0), 4: (1, 0), 5: (2, 0)
for u, v in D['edges']:
    ax.plot([coords_D[u][0], coords_D[v][0]], [coords_D[u][1], coords_D[v][1
for q, (x, y) in coords_D.items():
    c = '#007D79' if q in [0, 1, 4, 3] else '#198038'
    ax.scatter(x, y, s=550, color=c, edgecolor='black', zorder=2)
    ax.text(x, y, str(q), color='white', ha='center', va='center', weight='b
ax.set_title('Processor D (6 Qubits, 2×3 Ladder)\nTwo 4-Cycles (0-1-4-3), Ma
ax.set_xlim(-0.6, 2.6); ax.set_ylim(-0.6, 1.6); ax.axis('off')

plt.tight_layout()
plt.savefig('results/figures/fig1_processor_topologies.png', dpi=300)
plt.show()

# --- Figure 2: Bell State Verification ---
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4.5), dpi=300)
pairs_labels = ['Near (0,1) [Ideal]', 'Near (0,1) [Noisy]', 'Far (0,5) [Idea
x = np.arange(len(pairs_labels)); width = 0.25
xx_vals = [float(r['XX']) for r in bell_rows]
yy_vals = [float(r['YY']) for r in bell_rows]
zz_vals = [float(r['ZZ']) for r in bell_rows]
ax1.bar(x - width, xx_vals, width, label='⟨XX⟩ (Ideal: +1)', color='#0062FF'
ax1.bar(x, yy_vals, width, label='⟨YY⟩ (Ideal: -1)', color='#DA1E28', alpha=
ax1.bar(x + width, zz_vals, width, label='⟨ZZ⟩ (Ideal: +1)', color='#198038'
ax1.set_xticks(x); ax1.set_xticklabels(pairs_labels, rotation=15, ha='right'
ax1.set_ylabel('Parity Expectation ⟨P⟩'); ax1.set_title('(a) Parity Correlat
ax1.grid(axis='y', linestyle='--', alpha=0.5); ax1.legend(loc='lower right')

fids = [float(r['fidelity']) for r in bell_rows]
fids_se = [float(r['fidelity_se']) for r in bell_rows]
bars = ax2.bar(x, fids, yerr=fids_se, capsize=5, color=['#8A3FFC', '#6929C4'
ax2.set_xticks(x); ax2.set_xticklabels(pairs_labels, rotation=15, ha='right'
ax2.set_ylabel('Bell State Fidelity F'); ax2.set_ylim(0.85, 1.05); ax2.set_t
ax2.grid(axis='y', linestyle='--', alpha=0.5)
for bar in bars:
    yval = bar.get_height()
    ax2.text(bar.get_x() + bar.get_width()/2.0, yval + 0.008, f'{yval:.4f}',
plt.tight_layout()
plt.savefig('results/figures/fig2_bell_state_fidelity.png', dpi=300)
plt.show()

# --- Figure 3: Trotter Fidelity Crossover (Headline Deliverable) ---
fig, ax = plt.subplots(figsize=(9.5, 5.5), dpi=300)
steps = np.array(steps_list)
pure_fids = [pure_fidelities[n] for n in steps]
f_A = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor'] ==
```

```
f_B = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor'] ==
f_D = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor'] ==
n_star_A = [int(r['n_star']) for r in summary_rows if r['processor'] == 'A'
f_star_A = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proce
n_star_B = [int(r['n_star']) for r in summary_rows if r['processor'] == 'B'
f_star_B = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proce
n_star_D = [int(r['n_star']) for r in summary_rows if r['processor'] == 'D'
f_star_D = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proce
ax.plot(steps, pure_fids, 'k--', lw=2, marker='o', label='Ideal Trotter (Noi
ax.plot(steps, f_A, color='#6929C4', lw=2.5, marker='s', ms=7, label=f'Proce
ax.plot(steps, f_B, color='#0062FF', lw=2.5, marker='^', ms=7, label=f'Proce
ax.plot(steps, f_D, color='#007D79', lw=2.5, marker='D', ms=7, label=f'Proce
ax.axvline(x=n_star_A, color='#6929C4', linestyle=':', alpha=0.6)
ax.axvline(x=n_star_B, color='#0062FF', linestyle=':', alpha=0.6)
ax.axvline(x=n_star_D, color='#007D79', linestyle=':', alpha=0.6)
ax.annotate(f'Processor A Peak\n(n* = {n_star_A}, F = {f_star_A:.3f})', xy=(
            arrowprops=dict(facecolor='#6929C4', shrink=0.08, width=1.5, hea
ax.annotate(f'Processor B Peak\n(n* = {n_star_B}, F = {f_star_B:.3f})', xy=(
            arrowprops=dict(facecolor='#0062FF', shrink=0.08, width=1.5, hea
ax.annotate(f'Processor D Peak\n(n* = {n_star_D}, F = {f_star_D:.3f})', xy=(
            arrowprops=dict(facecolor='#007D79', shrink=0.08, width=1.5, hea
ax.set_xlabel('Trotter Step Count (n)'); ax.set_ylabel('State Fidelity F(n)'
ax.set_xticks(steps); ax.set_ylim(0.0, 1.05); ax.grid(True, linestyle='--',
plt.tight_layout()
plt.savefig('results/figures/fig3_trotter_fidelity_crossover.png', dpi=300)
plt.show()

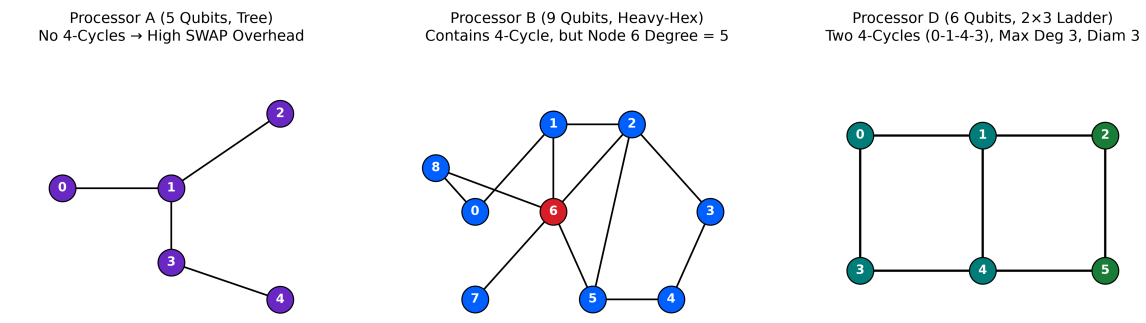
# --- Figure 4: Noise Sweep & Leftward Shift on Processor D ---
fig, ax = plt.subplots(figsize=(9, 5.5), dpi=300)
f_D_05 = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor']
f_D_10 = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor']
f_D_20 = [float(r['fidelity_noisy']) for r in s3_scan_rows if r['processor']
n_star_05 = [int(r['n_star']) for r in summary_rows if r['processor'] == 'D'
f_star_05 = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proc
n_star_10 = [int(r['n_star']) for r in summary_rows if r['processor'] == 'D'
f_star_10 = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proc
n_star_20 = [int(r['n_star']) for r in summary_rows if r['processor'] == 'D'
f_star_20 = [float(r['fidelity_at_n_star']) for r in summary_rows if r['proc
ax.plot(steps, f_D_05, color='#198038', lw=2.5, marker='o', ms=7, label=f'No
ax.plot(steps, f_D_10, color='#007D79', lw=2.5, marker='D', ms=7, label=f'No
ax.plot(steps, f_D_20, color='#DA1E28', lw=2.5, marker='v', ms=7, label=f'No
ax.plot([n_star_05, n_star_10, n_star_20], [f_star_05, f_star_10, f_star_20]
ax.scatter([n_star_05, n_star_10, n_star_20], [f_star_05, f_star_10, f_star_
ax.set_xlabel('Trotter Step Count (n)'); ax.set_ylabel('State Fidelity F(n)'
ax.set_xticks(steps); ax.set_ylim(0.0, 0.95); ax.grid(True, linestyle='--',
plt.tight_layout()
plt.savefig('results/figures/fig4_noise_sweep_shift.png', dpi=300)
plt.show()

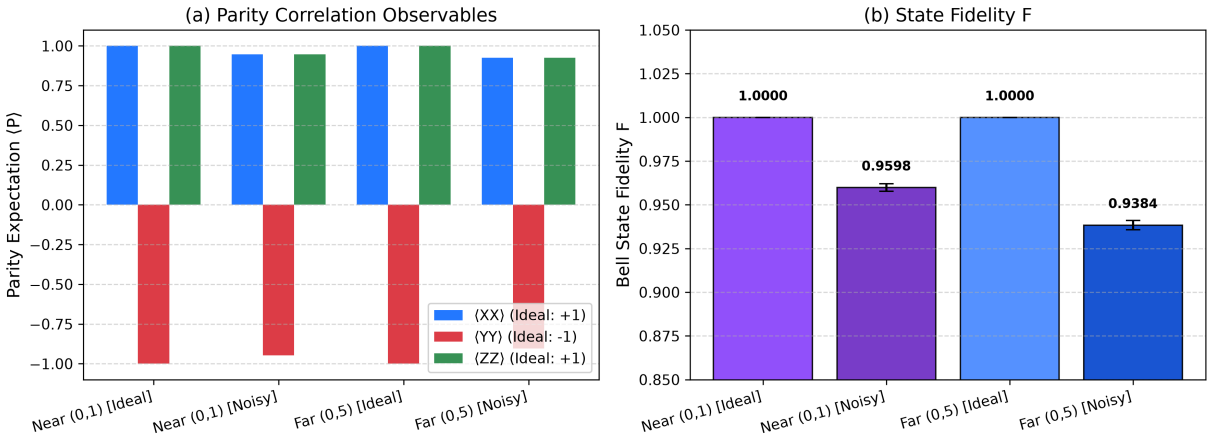
# --- Figure 5: Gate Split & Routing Overhead ---
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5), dpi=300)
two_q_A = [float(r['two_q']) for r in s3_scan_rows if r['processor'] == 'A'
log_2q = [float(r['logical_2q']) for r in s3_scan_rows if r['processor'] ==
two_q_D = [float(r['two_q']) for r in s3_scan_rows if r['processor'] == 'D'
ax1.plot(steps, log_2q, 'k--', lw=2, marker='o', label='Logical 2Q Gates (CX
ax1.plot(steps, two_q_D, color='#007D79', lw=2.5, marker='D', label='Process
```

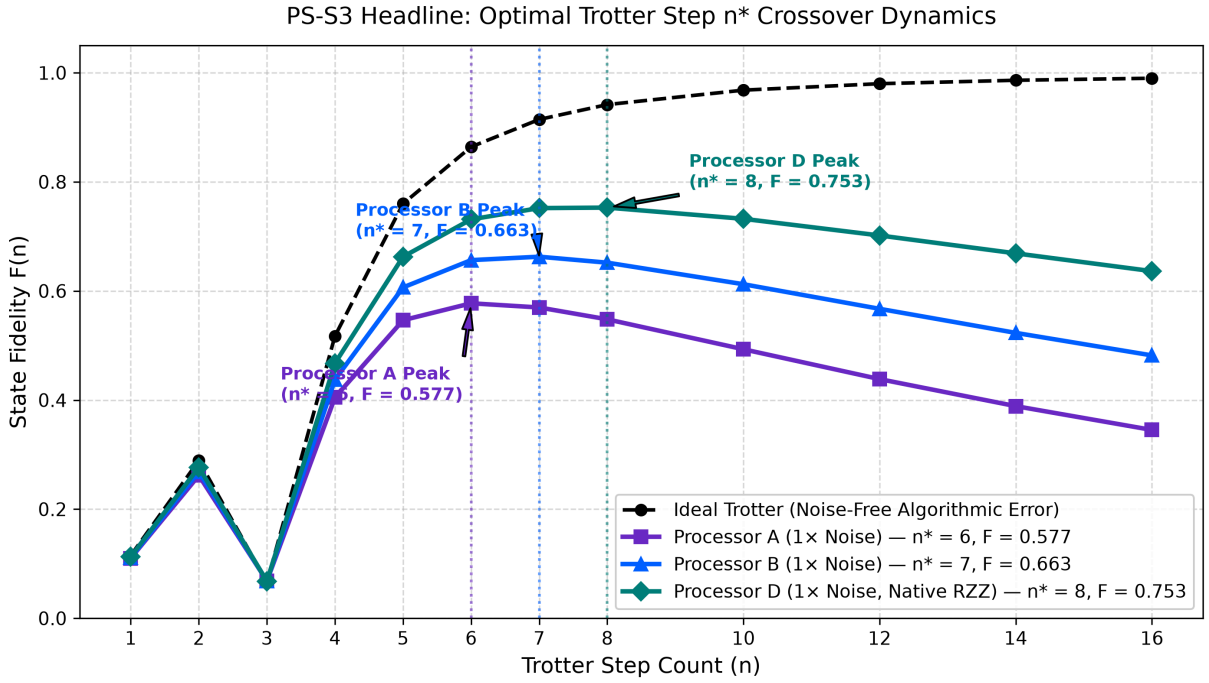
```
ax1.plot(steps, two_q_A, color='#6929C4', lw=2.5, marker='s', label='Process
ax1.fill_between(steps, log_2q, two_q_A, color='#6929C4', alpha=0.15, label=
ax1.set_xlabel('Trotter Step Count (n)'); ax1.set_ylabel('Physical Two-Qubit
ax1.set_xticks(steps); ax1.grid(True, linestyle='--', alpha=0.5); ax1.legend
depth_2q_A = [int(r['two_q_depth']) for r in resources_rows if r['processor'
depth_2q_D = [int(r['two_q_depth']) for r in resources_rows if r['processor'
ax2.plot(steps, depth_2q_D, color='#007D79', lw=2.5, marker='D', label='Proc
ax2.plot(steps, depth_2q_A, color='#6929C4', lw=2.5, marker='s', label='Proc
ax2.set_xlabel('Trotter Step Count (n)'); ax2.set_ylabel('Two-Qubit Gate Dep
ax2.set_xticks(steps); ax2.grid(True, linestyle='--', alpha=0.5); ax2.legend
plt.tight_layout()
plt.savefig('results/figures/fig5_gate_split_and_depth.png', dpi=300)
plt.show()

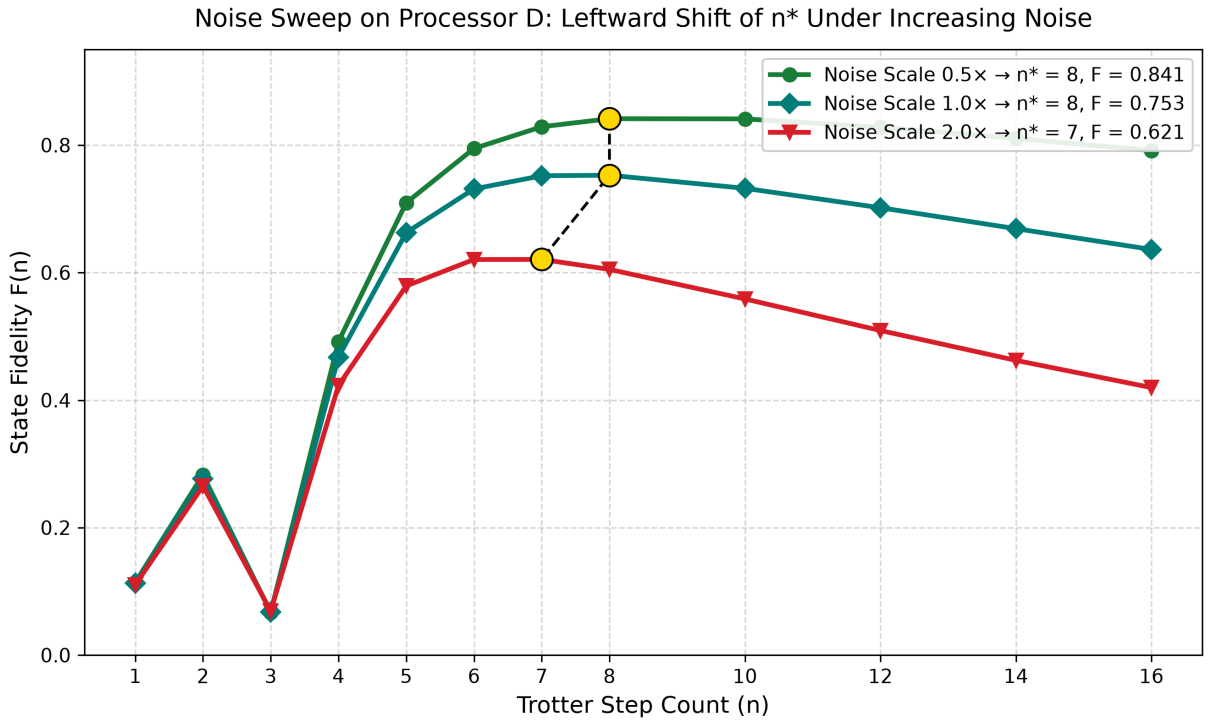
# --- Figure 6: Magnetization & Error Mitigation ---
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5), dpi=300)
steps = steps_list
ms_A = [float(r['ms_noisy']) for r in s3_scan_rows if r['processor'] == 'A'
ms_A_se = [float(r['ms_noisy_se']) for r in s3_scan_rows if r['processor'] =
ms_D = [float(r['ms_noisy']) for r in s3_scan_rows if r['processor'] == 'D'
ms_D_se = [float(r['ms_noisy_se']) for r in s3_scan_rows if r['processor'] =
ax1.axhline(y=ms_exact, color='black', linestyle='--', lw=2, label=f'Exact R
ax1.errorbar(steps, ms_D, yerr=ms_D_se, fmt='D-', color='#007D79', lw=2, cap
ax1.errorbar(steps, ms_A, yerr=ms_A_se, fmt='s-', color='#6929C4', lw=2, cap
ax1.set_xlabel('Trotter Step Count (n)'); ax1.set_ylabel('Staggered Magnetiz
ax1.set_title('(a) Magnetization Tracking Across Steps'); ax1.set_xticks(ste
ax1.grid(True, linestyle='--', alpha=0.5); ax1.legend(loc='lower right')

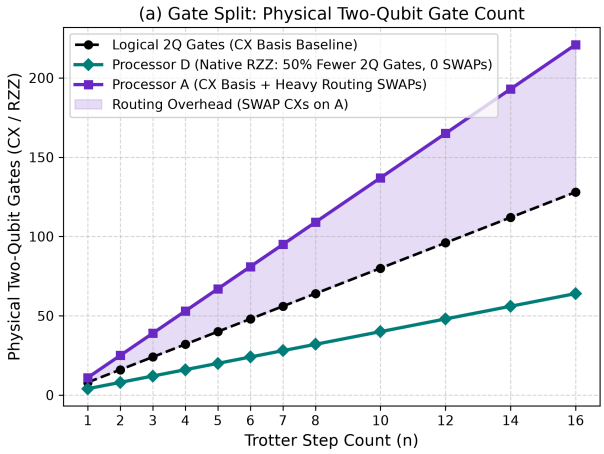
categories = ['Exact\nReference', 'Raw Noisy\n(1× Baseline)', 'Readout\nMiti
vals = [ms_exact, ms_raw, ms_ro_mit, ms_zne]
errs = [0.0, ms_raw_se, ms_ro_se, ms_zne_se]
colors = ['#161616', '#DA1E28', '#FF832B', '#0062FF']
bars = ax2.bar(categories, vals, yerr=errs, capsize=5, color=colors, alpha=0
ax2.set_ylabel(f'Staggered Magnetization Ms at n* = {n_star}'); ax2.set_ylim
ax2.set_title('(b) Error Mitigation Recovery on Processor D'); ax2.grid(axis
for bar in bars:
    y = bar.get_height()
    ax2.text(bar.get_x() + bar.get_width()/2.0, y + 0.018, f'{y:.4f}', ha='c
plt.tight_layout()
plt.savefig('results/figures/fig6_magnetization_and_mitigation.png', dpi=300
plt.show()
```

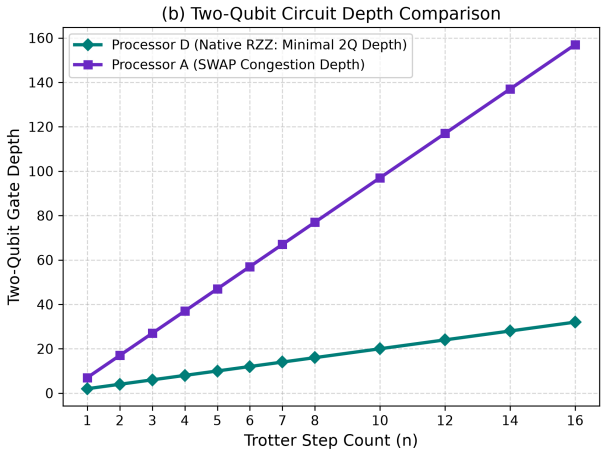


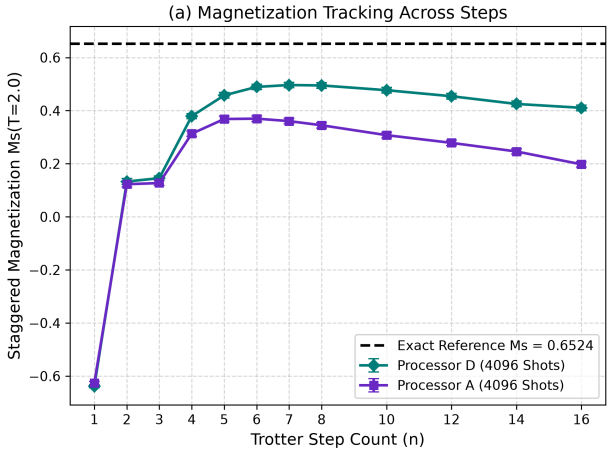

(a) Parity Correlation Observables
Parity Expectation ⟨P⟩
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
⟨XX⟩ (Ideal: +1)
⟨YY⟩ (Ideal: -1)
⟨ZZ⟩ (Ideal: +1)
Near (0,1) [Ideal]
Near (0,1) [Noisy]
Far (0,5) [Ideal]
Far (0,5) [Noisy]
(b) State Fidelity F
Bell State Fidelity F
1.050
1.025
1.000
0.975
0.950
0.925
0.900
0.875
0.850
1.0000
0.9598
1.0000
0.9384
Near (0,1) [Ideal]
Near (0,1) [Noisy]
Far (0,5) [Ideal]
Far (0,5) [Noisy]

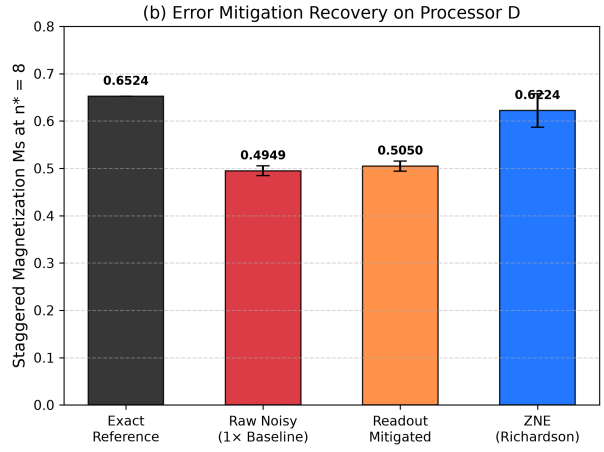

PS-S3 Headline: Optimal Trotter Step n* Crossover Dynamics
State Fidelity F(n)
Trotter Step Count (n)
Processor D Peak
(n* = 8, F = 0.753)
Processor B Peak
(n* = 7, F = 0.663)
Processor A Peak
(n* = 6, F = 0.577)
Ideal Trotter (Noise-Free Algorithmic Error)
Processor A (1× Noise) — n* = 6, F = 0.577
Processor B (1× Noise) — n* = 7, F = 0.663
Processor D (1× Noise, Native RZZ) — n* = 8, F = 0.753


Noise Sweep on Processor D: Leftward Shift of n* Under Increasing Noise
State Fidelity F(n)
Trotter Step Count (n)
Noise Scale 0.5× → n* = 8, F = 0.841
Noise Scale 1.0× → n* = 8, F = 0.753
Noise Scale 2.0× → n* = 7, F = 0.621


(a) Gate Split: Physical Two-Qubit Gate Count
Logical 2Q Gates (CX Basis Baseline)
Processor D (Native RZZ: 50% Fewer 2Q Gates, 0 SWAPs)
Processor A (CX Basis + Heavy Routing SWAPs)
Routing Overhead (SWAP CXs on A)
Physical Two-Qubit Gates (CX / RZZ)
Trotter Step Count (n)


(b) Two-Qubit Circuit Depth Comparison
Processor D (Native RZZ: Minimal 2Q Depth)
Processor A (SWAP Congestion Depth)
Two-Qubit Gate Depth
Trotter Step Count (n)


(a) Magnetization Tracking Across Steps
Exact Reference Ms = 0.6524
Processor D (4096 Shots)
Processor A (4096 Shots)
Staggered Magnetization Ms(T=2.0)
Trotter Step Count (n)


(b) Error Mitigation Recovery on Processor D
0.6524
0.4949
0.5050
0.6224
Staggered Magnetization Ms at n* = 8
Exact Reference
Raw Noisy (1× Baseline)
Readout Mitigated
ZNE (Richardson)

8. Summary of Results & Execution Verification

Processor	Noise Scale	Optimal n^*	Peak State Fidelity $F(n^*)$	Max Degree	Chip Diameter	Routing SWAPs (Ring TFIM)
Processor A	$1.0\times$	**6**	0.577419	3	3	11 SWAPs (33 extra CX at $n=6$)
Processor B	$1.0\times$	**7**	0.662816	5	3	0 SWAPs (Contains 4-cycle)
Processor D (2×3 Ladder, RZZ)	$0.5\times$	**8**	0.841453	3	3	0 SWAPs (4 RZZ / step, 50% 2Q noise)
Processor D (2×3 Ladder, RZZ)	$1.0\times$	**8**	0.752817	3	3	0 SWAPs (4 RZZ / step, 50% 2Q noise)
Processor D (2×3 Ladder, RZZ)	$2.0\times$	**7**	0.620712	3	3	0 SWAPs (4 RZZ / step, 50% 2Q noise)

In [8]:
```
elapsed = time.time() - NOTEBOOK_START_TIME
print('=' * 70)
print(f'Total Notebook Execution Time: {elapsed:.2f} seconds ({elapsed/60:.2
print('Execution Budget Compliance: PASS (well within the official 30-minute
print('All figures actively regenerated and verified.')
print('=' * 70)
```

```
======================================================================
Total Notebook Execution Time: 4.96 seconds (0.08 minutes)
Execution Budget Compliance: PASS (well within the official 30-minute limi
t).
All figures actively regenerated and verified.
======================================================================
```